

JAFP-21-0082  
September 8, 2021

Timothy C. Peter  
Site Vice President- JAF

United States Nuclear Regulatory Commission  
Attn: Document Control Desk  
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant  
Renewed Facility Operating License No. DPR-059  
NRC Docket No. 50-333

Subject: LER: 2020-003-01, High Pressure Coolant Injection Inoperable due to Oil Leak

Reference: FitzPatrick letter, LER: 2020-003, High Pressure Coolant Injection Inoperable due to Oil Leak, JAFP-20-0042, dated June 9, 2020

Dear Sir or Madam:

This report is supplement 1 to Licensee Event Report (LER) submitted by the Reference. This report is being submitted pursuant to 50.73(a)(2)(v)(D) and 50.73(a)(2)(i)(B). This LER revision is being submitted to include reference to reporting criterion 50.73(a)(2)(i)(B) for Operation or Condition Prohibited by Technical Specifications (TS) and include more information in the Event Analysis section. Subsequent analysis of the event determined the High Pressure Coolant Injection (HPCI) system was inoperable for a longer period than allowed by Technical Specifications 3.5.1.

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Richard Sullivan, Regulatory Assurance Manager, at (315) 349-6562.

Sincerely,



Timothy C. Peter  
Site Vice President

TCP/RS

Enclosure: LER: 2020-003-01, High Pressure Coolant Injection Inoperable due to Oil Leak

cc: USNRC, Region I Administrator  
USNRC, Project Manager  
USNRC, Resident Inspector  
INPO Records Center (IRIS)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------|--------------------------|-------------------------------------------------------------|-------------------------------------|--------------------|
| <b>NRC FORM 366</b><br>(08-2020)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        | <b>U.S. NUCLEAR REGULATORY COMMISSION</b>                                                                                                                                                                                                                                                                                                               |                      |                                                       | <b>APPROVED BY OMB: NO. 3150-0104</b> <b>EXPIRES: 08/31/2023</b> |                                                       |                          |                                                             |                                     |                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        | <b>LICENSEE EVENT REPORT (LER)</b><br>(See Page 3 for required number of digits/characters for each block)<br>(See NUREG-1022, R.3 for instruction and guidance for completing this form<br><a href="http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/">http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/</a> ) |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| <b>1. Facility Name</b><br>James A. FitzPatrick Nuclear Power Plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       | <b>2. Docket Number</b><br>05000333                              |                                                       | <b>3. Page</b><br>1 OF 3 |                                                             |                                     |                    |
| <b>4. Title</b><br>High Pressure Coolant Injection Inoperable due to Oil Leak                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| <b>5. Event Date</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                                                                         | <b>6. LER Number</b> |                                                       |                                                                  | <b>7. Report Date</b>                                 |                          |                                                             | <b>8. Other Facilities Involved</b> |                    |
| Month                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Day    | Year                                                                                                                                                                                                                                                                                                                                                    | Year                 | Sequential Number                                     | Revision No.                                                     | Month                                                 | Day                      | Year                                                        | Facility Name                       | Docket Number      |
| 04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10     | 2020                                                                                                                                                                                                                                                                                                                                                    | 2020 -               | 003                                                   | - 01                                                             | 09                                                    | 08                       | 2021                                                        | N/A                                 | N/A                |
| <b>9. Operating Mode</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  | <b>10. Power Level</b>                                |                          |                                                             |                                     |                    |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  | 100                                                   |                          |                                                             |                                     |                    |
| <b>11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| <b>10 CFR Part 20</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        | <input type="checkbox"/> 20.2203(a)(2)(vi)                                                                                                                                                                                                                                                                                                              |                      | <input type="checkbox"/> 50.36(c)(2)                  |                                                                  | <input type="checkbox"/> 50.73(a)(2)(iv)(A)           |                          | <input type="checkbox"/> 50.73(a)(2)(x)                     |                                     |                    |
| <input type="checkbox"/> 20.2201(b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        | <input type="checkbox"/> 20.2203(a)(3)(i)                                                                                                                                                                                                                                                                                                               |                      | <input type="checkbox"/> 50.46(a)(3)(ii)              |                                                                  | <input type="checkbox"/> 50.73(a)(2)(v)(A)            |                          | <b>10 CFR Part 73</b>                                       |                                     |                    |
| <input type="checkbox"/> 20.2201(d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        | <input type="checkbox"/> 20.2203(a)(3)(ii)                                                                                                                                                                                                                                                                                                              |                      | <input type="checkbox"/> 50.69(g)                     |                                                                  | <input type="checkbox"/> 50.73(a)(2)(v)(B)            |                          | <input type="checkbox"/> 73.71(a)(4)                        |                                     |                    |
| <input type="checkbox"/> 20.2203(a)(1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        | <input type="checkbox"/> 20.2203(a)(4)                                                                                                                                                                                                                                                                                                                  |                      | <input type="checkbox"/> 50.73(a)(2)(i)(A)            |                                                                  | <input type="checkbox"/> 50.73(a)(2)(v)(C)            |                          | <input type="checkbox"/> 73.71(a)(5)                        |                                     |                    |
| <input type="checkbox"/> 20.2203(a)(2)(i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        | <b>10 CFR Part 21</b>                                                                                                                                                                                                                                                                                                                                   |                      | <input checked="" type="checkbox"/> 50.73(a)(2)(i)(B) |                                                                  | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(D) |                          | <input type="checkbox"/> 73.77(a)(1)                        |                                     |                    |
| <input type="checkbox"/> 20.2203(a)(2)(ii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        | <input type="checkbox"/> 21.2(c)                                                                                                                                                                                                                                                                                                                        |                      | <input type="checkbox"/> 50.73(a)(2)(i)(C)            |                                                                  | <input type="checkbox"/> 50.73(a)(2)(vii)             |                          | <input type="checkbox"/> 73.77(a)(2)(i)                     |                                     |                    |
| <input type="checkbox"/> 20.2203(a)(2)(iii)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        | <b>10 CFR Part 50</b>                                                                                                                                                                                                                                                                                                                                   |                      | <input type="checkbox"/> 50.73(a)(2)(ii)(A)           |                                                                  | <input type="checkbox"/> 50.73(a)(2)(viii)(A)         |                          | <input type="checkbox"/> 73.77(a)(2)(ii)                    |                                     |                    |
| <input type="checkbox"/> 20.2203(a)(2)(iv)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        | <input type="checkbox"/> 50.36(c)(1)(i)(A)                                                                                                                                                                                                                                                                                                              |                      | <input type="checkbox"/> 50.73(a)(2)(ii)(B)           |                                                                  | <input type="checkbox"/> 50.73(a)(2)(viii)(B)         |                          |                                                             |                                     |                    |
| <input type="checkbox"/> 20.2203(a)(2)(v)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        | <input type="checkbox"/> 50.36(c)(1)(ii)(A)                                                                                                                                                                                                                                                                                                             |                      | <input type="checkbox"/> 50.73(a)(2)(iii)             |                                                                  | <input type="checkbox"/> 50.73(a)(2)(ix)(A)           |                          |                                                             |                                     |                    |
| <input type="checkbox"/> OTHER (Specify in Abstract below or in NRC Form 366A).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| <b>12. Licensee Contact for this LER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| <b>Licensee Contact</b><br>Mr. Richard Sullivan, Regulatory Assurance Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          | <b>Telephone Number (Include Area Code)</b><br>315-349-6562 |                                     |                    |
| <b>13. Complete One Line for each Component Failure Described in this Report</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| Cause                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | System | Component                                                                                                                                                                                                                                                                                                                                               | Manufacture          | Reportable to IRIS                                    |                                                                  | Cause                                                 | System                   | Component                                                   | Manufacturer                        | Reportable to IRIS |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SJ     | PCV                                                                                                                                                                                                                                                                                                                                                     | R290                 | Y                                                     |                                                                  |                                                       |                          |                                                             |                                     |                    |
| <b>14. Supplemental Report Expected</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  | <b>15. Expected Submission Date</b>                   |                          |                                                             |                                     |                    |
| <input checked="" type="checkbox"/> No <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission date)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  | Month      Day      Year                              |                          |                                                             |                                     |                    |
| <b>Abstract</b> (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |
| On April 10, 2020 at 0300, an oil leak from 23PCV-12, HPCI Trip System Pressure Control Valve (PCV) resulted in the system being declared inoperable. This condition is being reported as a condition that could have prevented the fulfillment of the safety function of a system needed to mitigate the consequences of an accident per 10 CFR 50.73(a)(2)(v)(D) and for a period longer than allowed by Technical Specification 3.5.1 per 10 CFR 50.73(a)(2)(i)(B).<br><br>Corrective action to replace valve 23PCV-12 was completed and system was restored to service at time 2015 on April 10, 2020. |        |                                                                                                                                                                                                                                                                                                                                                         |                      |                                                       |                                                                  |                                                       |                          |                                                             |                                     |                    |

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: [oira\\_submission@omb.eop.gov](mailto:oira_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

| 1. FACILITY NAME                         | 2. DOCKET NUMBER | 3. LER NUMBER |                   |         |
|------------------------------------------|------------------|---------------|-------------------|---------|
|                                          |                  | YEAR          | SEQUENTIAL NUMBER | REV NO. |
| James A. FitzPatrick Nuclear Power Plant | 05000 – 333      | 2020          | – 003             | – 01    |

**NARRATIVE****Background**

The High Pressure Coolant Injection (HPCI) System [EIS Identifier: BJ] is part of the Emergency Core Cooling System (ECCS); HPCI provides and maintains an adequate coolant inventory inside the Reactor Pressure Vessel [RPV] to prevent damage to the reactor core under postulated accident scenarios. The system is comprised of various components which include a turbine, pumps, valves, piping, and instrumentation. The HPCI System may be initiated by automatic or manual means.

The HPCI turbine oil system provides for lubrication and cooling of the equipment bearings and other accessories. The oil system also provides control and hydraulic oil to the turbine governor system. Loss of oil pressure to the turbine governor system will cause the steam supply stop valve and turbine control valve to go shut, securing steam to the turbine. The HPCI oil system is not normally pressurized, the system is only pressurized during operational checks of the system and during actual system operation.

**Event Description**

While conducting a HPCI monthly operability test (ST-4B), a control oil leak was observed from 23PCV-12. The oil leak is only visible when the oil system is pressurized. Initial leakage rate was estimated at approximately 0.1 gallons per minute. To confirm leak location and validate leak rate, the auxiliary oil pump was restarted. With system oil pressure reestablished, the leak rate was determined to have degraded such that the capability of the HPCI system to meet mission time would be challenged.

**Event Analysis**

Level in the HPCI Oil Tank provides net positive suction head (NPSH) to the suctions of the oil pumps for the duration of HPCI mission time. Loss of NPSH to the suction of HPCI Main Lube Oil Pump (23P-1MO) and the HPCI Turbine Aux. Lube Oil Pump (23P-150) will result in a loss of oil pressure to the system bearings and system control valves. Oil pressure is required for operation of the HPCI Turbine Steam Supply Stop Valve (23HOV-1) and HPCI Turbine Control Valve (23HOV-2). Loss of bearing cooling and oil pressure for the stop and control valves would have limited HPCI operation and challenged the capability of the HPCI system to meet its mission time. 23PCV-12 was installed on December 16, 2017. It was identified that this PCV had a vulnerability associated with the HPCI system oil PCV actuator diaphragm due to a manufacturing error, as reported by General Electric-Hitachi MFN 10-192. This error resulted in inadequate fabric reinforcement that is critical to ensure durability and reliability of the diaphragm, preventing tearing of the diaphragm. The failure of 23PCV-12 diaphragm would result in a loss of HPCI system turbine lubricating and control oil through the failed diaphragm. This condition is reportable under 10 CFR 50.73(a)(2)(v)(D), Event or Condition that Could Have Prevented Fulfillment of a Safety Function.

Due to the cyclic failure mechanism of the diaphragm, HPCI was inoperable longer than allowed by Technical Specifications (TS) 3.5.1, ECCS Operating. TS 3.5.1 Condition C requires HPCI System be restored to Operable within 14 days or, Condition G, be in Mode 3 and reduce reactor steam dome pressure to  $\leq 150$  psig within 12 hours. This represents a condition prohibited by Tech Spec and reportable in accordance with 10 CFR 50.73(a)(2)(i)(B).

**Cause**

The cause was determined to be a tear in the 23PCV-12 control valve diaphragm, allowing a loss of control oil from the system.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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**Similar Events**

Previous LER: None identified.

**FAILED COMPONENT IDENTIFICATION:**

|                            |            |
|----------------------------|------------|
| Manufacturer:              | Robertshaw |
| Manufacturer Model Number: | VC-210     |
| NPRDS Manufacturer Code:   | R290       |
| NPRDS Component Code:      | PCV        |
| FitzPatrick Component ID:  | 23PCV-12   |

**Completed Corrective Actions**

Valve 23PCV-12 was replaced. Additional corrective actions are documented in the Corrective Action Program.

**Safety Significance**

**Nuclear Safety** - The failure of the HPCI turbine over-speed reset control valve diaphragm would result in a loss of HPCI turbine lubricating oil and control oil during HPIC oil system operation. Depending on the amount of oil lost and system demands, the leakage may have prevented the HPCI from meeting system specified mission time.

**Industrial safety** – No actual or potential industrial safety consequences resulted from this event.

**Radiological safety** – No actual radiological impact is associated with this event.

**References**

- Issue Report – IR 04334315, HPCI Oil Leak, dated April 10, 2020
- Issue Report – IR 04397491, NRC NCV 05000333/2020012-01, Defective Part HPIC PCV Failure, dated January 22, 2021
- ENS 54657, issued 4/10/2020
- Part 21 Reportable Condition Notification: Failure of HPCI Turbine Overspeed Reset Control Valve Diaphragm, MFN 10-192, dated July 1, 2010